Software Requirements Specification

for

< Smart Attendance System >

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

This document describes the requirements of the Smart Attendance System. The system automatically records attendance using face recognition technology. It reduces manual work and improves accuracy in attendance tracking. The document includes both functional and non-functional requirements. It also acts as a reference for development and testing.

1.2 Document Conventions

All requirements are written using a structured format with proper numbering. Requirement IDs such as REQ-1, REQ-2 are used for identification. Important terms are highlighted clearly for easy understanding. Simple and consistent language is followed throughout the document. This helps all stakeholders easily understand the requirements.

1.3 Intended Audience

This document is intended for developers, testers, and system users. Developers use it to build the system based on requirements. Testers use it to verify and validate system functions. Faculty and administrators use it to understand system operations. It helps all users get a clear idea of the system.

1.4 Product Scope

The system is designed to automate attendance using face recognition technology. It captures student images and compares them with stored data. It eliminates manual errors and saves time in attendance marking. The system also generates reports for analysis. It can be used in schools, colleges, and offices.

1.5 References

This document follows the IEEE SRS standard template for documentation. Software Engineering textbooks and notes are referred for structure. Face recognition concepts are taken from Artificial Intelligence resources. Online tutorials and academic materials are also used. These references ensure accuracy and completeness of the document.

2. Overall Description

2.1 Product Perspective

The Smart Attendance System is a standalone application designed to replace manual attendance systems. It works along with camera hardware to capture images. The

system is part of modern smart campus solutions. It interacts with a database for storing and retrieving records. It can be expanded in the future with additional features.

2.2 Product Functions

The system captures images using a camera and detects faces. It recognizes individuals by comparing with stored data. Attendance is marked automatically after successful recognition. It also generates attendance reports for users. The system ensures accuracy and prevents duplicate entries.

2.3 User Classes and Characteristics

There are different types of users such as admin, faculty, and students. Admin has full control over system operations and data management. Faculty can view attendance reports and monitor students. Students can check their attendance status. Users are expected to have basic computer knowledge.

2.4 Operating Environment

The system operates on platforms like Windows or Linux. It requires a webcam or camera for capturing images. Programming tools such as Python and OpenCV are used. A database like MySQL is needed to store data. The system works on standard computer hardware.

2.5 Design and Implementation Constraints

The system requires proper lighting conditions for accurate face detection. Camera quality affects recognition performance. Hardware limitations may impact system speed. Security measures must be implemented to protect data. The system must be compatible with available software tools.

2.6 User Documentation

User manuals will be provided to guide users in operating the system. Help sections will be included within the application. Training materials may be given to faculty and staff. Instructions will be simple and easy to follow. Documentation will be updated when the system is improved.

2.7 Assumptions and Dependencies

It is assumed that all users are registered in the system database. The camera hardware is expected to function properly. Users should follow system instructions correctly. Internet may be required for certain features like cloud storage. The system depends on accurate face recognition technology.

3. External Interface Requirements

3.1 User Interfaces

The system provides a simple and user-friendly interface. It includes login pages, dashboards, and report screens. Navigation is easy with clearly labeled buttons and menus. Error messages are displayed properly for user understanding. The interface is designed to be attractive and easy to use.

3.2 Hardware Interfaces

The system interacts with hardware components such as a webcam or camera. The camera captures images for face detection. A computer system is required to process the data. Hardware devices must be compatible with the software. Proper power supply is needed for smooth operation.

3.3 Software Interfaces

The system uses a database like MySQL for storing attendance data. Face recognition libraries are integrated into the system. The operating system supports execution of the application. Different software modules interact internally. All components must work together without conflicts.

3.4 Communication Interfaces

The system may use internet connectivity for data transfer. Communication protocols such as HTTP can be used. Data can be stored or accessed from cloud servers. Secure communication is important to protect data. The network should support reliable and fast communication.

4. System Features

4.1 Face Detection Module

4.1.1 Description and Priority

This module is responsible for detecting faces using a camera. It is the main component of the system and ensures accurate identification. The feature works in real-time conditions. It plays a critical role in the overall system performance. Hence, it is assigned high priority.

4.1.2 Stimulus/Response Sequences

When a user stands in front of the camera, the system captures the image. The face is detected and processed for recognition. The system compares the face with stored data. If a match is found, it proceeds further. If not, an error message is displayed.

4.1.3 Functional Requirements

The system shall detect faces using the camera. It shall capture images automatically without user input. The system shall compare detected faces with the database. It shall identify valid users correctly. It shall handle errors for unrecognized faces.

4.2 Attendance Marking

4.2.1 Description and Priority

This module marks attendance automatically after successful face recognition. It stores attendance details in the database. The system ensures that attendance is recorded only once. It reduces manual errors and improves efficiency. This feature is also given high priority.

4.2.2 Stimulus/Response Sequences

After recognizing a face, the system checks attendance status. If attendance is not marked, it records the entry. If already marked, it avoids duplication. The system stores date and time of attendance. A confirmation message is displayed to the user.

4.2.3 Functional Requirements

The system shall mark attendance automatically. It shall store date and time details. It shall prevent duplicate attendance entries. The system shall update records in the database. It shall generate attendance reports when required.

5. Non-Functional Requirements

5.1 Performance Requirements

The system should respond quickly within a few seconds. Face detection and recognition must be fast and accurate. Database operations should be efficient. The system should support multiple users at the same time. Performance should remain stable under normal conditions.

5.2 Safety Requirements

The system should prevent loss of data through proper backup. Errors should be handled without affecting system performance. Incorrect attendance marking should be avoided.

Hardware failures should be managed safely. Safety measures must be implemented throughout the system.

5.3 Security Requirements

User authentication is required to access the system. Face data should be securely stored in the database. Unauthorized access must be prevented. Data encryption can be used for protection. Privacy of users must be maintained at all times.

5.4 Software Quality Attributes

The system should be reliable and easy to use. It must be maintainable for future updates. Accuracy of face recognition should be high. The system should be scalable for additional users. User satisfaction is an important quality factor.

5.5 Business Rules

Attendance is marked only during specific time periods. Each user can mark attendance only once per session. Admin controls all system operations. Reports must be generated regularly. The system must follow institutional rules and policies.

6. Other Requirements

The system should support future enhancements and updates. Mobile application integration can be added later. Cloud storage features may be implemented. Data backup and recovery should be available. The system should be flexible and scalable for future needs.

Appendix A: Glossary

AI refers to Artificial Intelligence used in the system. Database refers to structured data storage. Face recognition is the process of identifying a person using an image. Admin is the system controller. User refers to anyone using the system.

Appendix B: Analysis Models

DFD diagrams can be used to represent data flow. Use case diagrams show interactions between users and the system. ER diagrams define database structure. Flowcharts explain system processes clearly. These models help in better understanding of the system.

Appendix C: To Be Determined

Some features like mobile integration are not finalized yet. Advanced AI improvements are under development. Cloud storage implementation is pending. Certain requirements may change in the future. These items will be updated later.